Exploratory Data Analysis — Report

Project 2 · Data Analytics Internship · Decode Labs

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Position:	Data Analytics Intern	**Mode:**	Remote / Virtual
Internship Period:	27 Sep 2026 – 27 Oct 2026	**Report Date:**	06 Oct 2026
Dataset:	Orders_Clean (cleaned in Project 1)	**Tool Used:**	Excel (PivotTables, formulas)

Objective

Analyze the cleaned Project 1 dataset to understand patterns, trends, and distributions, per the Project 2 brief: calculate basic statistics (mean, median, count), identify trends and outliers, and summarize key observations.

1. Descriptive Statistics

Metric	Quantity	UnitPrice	TotalPrice
Mean (Average)	2.95	356.41	1,053.97
Median	3.00	364.21	823.62
Count	1,200	1,200	1,200

The mean TotalPrice (1,053.97) sits approximately 28% above the median (823.62), indicating a right-skewed distribution — a small number of high-value orders are pulling the average upward, while most orders fall below it. Quantity and UnitPrice show mean and median close together, indicating a more balanced spread.

2. Outlier Analysis (IQR Method)

Metric	Value
Q1 (25th percentile)	410.52
Q3 (75th percentile)	1,578.48
IQR	1,167.96
Lower Limit	N/A (negative)
Upper Limit	3,330.41
Number of Outliers	8 (0.67% of dataset)

All 8 outliers in TotalPrice involve the maximum order quantity (5 units) of mid-to-high-priced products (Laptop, Printer, Tablet, Monitor, Chair). No data entry errors were found on inspection — these are legitimate bulk orders (signal, not noise) and directly explain the gap between the mean and median TotalPrice above.

3. Trend Analysis

By Product

Product	Order Count	Avg. TotalPrice
Chair	178	1,098.99
Desk	170	985.06
Laptop	173	1,110.56
Monitor	163	1,077.62
Phone	156	972.58
Printer	181	1,080.73
Tablet	179	1,042.28

Order volume is fairly even across products (156–181 orders). Phone is weak on both dimensions — lowest order count (156) and lowest average order value (972.58) — worth investigating further. Laptop has the highest average order value (1,110.56) despite only moderate order volume.

By Referral Source

Referral Source	Order Count	Avg. TotalPrice
Email	250	1,047.23
Facebook	228	1,098.29
Google	241	1,039.18
Instagram	259	1,062.88
Referral	222	1,021.69

Order volume is fairly even across channels (222–259 orders). Facebook-referred customers show the highest average order value (1,098.29) despite moderate order volume, suggesting this channel may attract a higher-value customer segment worth further marketing investment. Instagram drives the most orders overall (259) but only mid-pack average value.

4. Correlation Analysis

Variable Pair	Correlation (r)
Quantity vs TotalPrice	0.615
Quantity vs ItemsInCart	0.650

Both pairs show a moderate positive correlation — related, but not strong enough to be treated as reliable predictors on their own. Other factors (such as UnitPrice variation across product categories) clearly also influence TotalPrice. Per standard practice, correlation is treated here as a clue for further investigation, not as evidence of a causal relationship.

5. Key Observations Summary

1. Order value is right-skewed: the mean (1,053.97) sits ~28% above the median (823.62), driven by a small number of high-value orders.

2. The skew is explained by 8 legitimate bulk orders (0.67% of the dataset), each involving the maximum order quantity (5 units) — not data errors.

3. Phone underperforms on both order volume and average order value, making it a candidate for a pricing or promotion review.

4. Laptop generates the highest average order value despite only moderate order volume.

5. Facebook-referred customers show the highest average order value among referral channels, suggesting a higher-value customer segment worth further marketing investment.

6. Quantity shows a moderate (not strong) positive relationship with both TotalPrice and ItemsInCart — a clue for further analysis, not a standalone predictor.

Methodology

All analysis was performed in Microsoft Excel on the Orders_Clean dataset produced in Project 1. Descriptive statistics (mean, median, count) were calculated using AVERAGE, MEDIAN, and COUNT formulas on a dedicated Statistics sheet. Outliers were identified using the IQR method (1.5× IQR beyond Q1/Q3), then individually inspected in the dataset to distinguish legitimate bulk orders (signal) from potential data errors (noise). Trends were identified using PivotTables grouping order count and average order value by Product and by ReferralSource. Correlation was calculated using the CORREL function between relevant numeric variable pairs.

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